Junjie Yu

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Education

2023–26*	**Ph.D.**, Environmental Sciences, The University of Manchester
2020–23	**M.S.**, Environmental Engineering, Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences
2016–20	**B.Eng.**, Environmental Engineering, China University of Mining and Technology-Beijing

Awards & Honors

2025.06	The University of Manchester Open Research Award 2025
2024.11	Travel funding to attend Transforming Urban Living with AI and Digital Twins
2023.09	Significant Contributor in City Brain Open Research - Open Data Computing Workshop on Urban Climate
2022.06	National Scholarship for Graduate Students (the highest honor for postgraduates in China, 3%)
2020.06	Excellent Graduates in Beijing (5%)
2019.06	Special-Class Award in Beijing Energy and Water Conservation Low Carbon Emission Reduction Social Practice and Technology Competition
2018.12	Second Prize in Beijing Chemistry Experiment Competition
2017.11	First Prize in National University Student Mathematics Competition

Tools & Software

Data sciences

Google Colab: AutoML for observational weather data.

obswx: A Python package for accessing observational meteorological data (PyPi | GitHub).

Climate modeling

pyclmuapp: Integration and Execution of Community Land Model Urban (CLMU) in a Containerized Environment (PyPi | GitHub).

CLMU-App: Enabling Operating System Independent Urban Climate Simulations (GitHub).

*Expected.

Featured Projects

Diffusion Generative Model-Based Precipitation Downscaling and Health Risk Assessment

Focus: Generative AI & Spatiotemporal Prediction

- Engineered robust data pipelines to ingest, clean, and harmonize massive multi-source meteorological datasets, including satellite remote sensing, ERA5 reanalysis, and CMIP6 future climate projections, alongside UK mortality records.
- Designed and trained a state-of-the-art **Diffusion Generative Model** using **PyTorch** to perform high-resolution spatiotemporal downscaling. Successfully mapped coarse-resolution climate inputs to highly accurate, localized precipitation fields.
- Developed a precipitation-mortality risk attribution framework for the UK. Leveraged the generated high-fidelity meteorological data to significantly reduce uncertainty in health risk identification and evaluate the downstream impacts of extreme weather events.

Optimization of Urban Roof Water-Based Cooling Technologies Using Machine Learning and Optimization Algorithms

PhD Project

- Developed a Fortran module for urban roof water-based cooling technology and coupled it with the Community Land Model Urban.
- Evaluated the potential of advanced machine learning methods for building surrogate models on small-sample datasets.
- Utilized the surrogate models, combined with multi-objective optimization algorithms, for optimization of water-based cooling system operational parameters.
- Systematically evaluated the energy savings and climate benefits.
- Project Stage Outcomes:
  - Manuscript completed.

Physics-Informed Deep Learning for Local Climate Zones (LCZ)-Based Urban Climate

Funder: NVIDIA Corporation (2 x NVIDIA RTX PRO 6000 Blackwell GPUs)
Role: Co-Investigator
Date: September 2025

- Assist the PI in developing proposal.
- Led the construction of a multi-city training dataset encompassing diverse Local Climate Zone types and atmospheric conditions.
- Designed and implemented a Physics-Informed Deep Learning model integrating LCZ physical encoding and physical constraints (e.g., surface energy balance).
- Accelerated physical feature computation using CuPy and trained physics-embedded neural networks using the PhysicsNeMo framework, optimizing model training efficiency.
- Responsible for fine-tuning pre-trained models in data-scarce scenarios and systematically evaluating model generalization performance across spatial and temporal dimensions.
- Project Stage Outcomes:
  - Xia, J., Ling, F., Li, Z., **Junjie Yu**, Zhang, H., Topping, D. O., Bai, L., & Zheng, Z. (2025). Learning Urban Climate Dynamics via Physics-Guided Urban Surface–Atmosphere Interac-

tions. Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025, Main Track).
  - **Junjie Yu**, Zheng, Z., Lindley, S., Zhao, L., Wang, C., Wu, Q., Li, L., Topping, D., Schreck, J. S., Gagne, D. J. & Oleson, K. Leveraging Automated Machine Learning (AutoML) for Urban Climate Emulation. (accepted by Big Data and Earth System)

Cloud-based Platform for Democratizing Urban Climate Modeling

Funder: Amazon Web Services ($5,000 Cloud Computing Credits)
Role: Co-Investigator
Date: July 2025 – July 2026

- Assist the PI in developing proposal.
- Developed a full-stack application for running urban climate simulations driven by ERA5 data, using FastAPI (Backend) and HTML/JavaScript (Frontend UI).
- Designed and implemented cloud deployment on AWS ECS-Fargate, achieving containerized services, Application Load Balancer setup, and secure network configuration.
- Integrated S3-based data storage and designed Zarr data storage optimization for efficient data access.
- Configured domain names, CNAME records, target groups, and health checks, optimizing system performance and troubleshooting network bottlenecks.
- Project Stage Outcomes:
  - Online Cloud Platform: `http://pyclmuapp.open-urbanclimate.com/`
  - Registered users during testing phase: 50, from the UK, USA, China, and Belgium.
  - Served thousands of simulations.

Development of Smart Urban Climate Adaptation Strategies Using Reinforcement Learning

PhD Project

- Developed a coupled simulation framework integrating "offline training – online evaluation" reinforcement learning with an urban climate model.
- Implemented integration of various reinforcement learning algorithms (including Q-Learning, DQN, SAC, and PPO) with the urban climate model, and rewrote the policy prediction module in Fortran to enhance coupling efficiency.
- Analyzed the potential impacts of Reinforcement Learning-based HVAC control strategies on local climates.
- Project Stage Outcomes:
  - **Junjie Yu**, John S Schreck, David John Gagne, Keith W Oleson, Jie Li, Yongtu Liang, Qi Liao, Mingfei Sun, David O Topping, Zhonghua Zheng. Reinforcement Learning (RL) Meets Urban Climate Modeling: Investigating the Efficacy and Impacts of RL-Based HVAC Control. ArXiv preprint (2025).
  - Project code open-sourced: `https://github.com/envdes/code_CLMU_HVAC_RL`

City Brain Open Research - Open Data Computing Workshop on Urban Climate

Significant Contributor

- Leveraged cloud computing services to optimize and accelerate the processing and analysis pipelines for large-scale CMIP6 climate ensembles.

- Empowered big data processing for climate extremes research by utilizing cloud-based platforms to handle massive multi-dimensional meteorological datasets.
- Project Stage Outcomes:
  - Kuang Wang, **Junjie Yu**, Z. L. & Zheng, Z. Leverage a Cloud-based Big Data Processing Platform for Climate Extremes Research. AGU Fall Meeting 2023 Abstract. Dec. 2023.

Development of Streamlined Execution Technology for the Urban Climate Model – CLMU

PhD Project

- Developed an operating system-independent containerized version of the Community Land Model Urban (CLMU).
- Developed Python-based tools interfacing with the containerized CLMU to facilitate the generation of urban surface and atmospheric forcing data.
- Enabled simulations under future climate change scenarios to explore their impacts and potential urban climate responses.
- Project Outcomes:
  - **Junjie Yu**, S. Yuan, S. Lindley, C. Jay, D.O. Topping, K.W. Oleson, Z.Z Zheng. Integration and Execution of Community Land Model Urban (CLMU) in a Containerized Environment. Environ. Model. Softw.
  - PyPI: `https://pypi.org/project/pyclmuapp/`
  - Project code open-sourced: `https://github.com/envdes/pyclmuapp`

UK Research and Innovation Impact Acceleration Account Scheme Research Proposal

Funder: UK Research and Innovation (£10,000)
Role: Research and Innovation Assistant
Duration: April 2025

- Developed deep learning models integrating urban physics and observational data, including running urban physics models to generate data and constructing deep learning models.
- Assisted in organizing an urban climate resilience workshop, including workshop website design, agenda planning, personnel scheduling, and chairing sessions.
- Project Stage Outcome: `https://envdes.github.io/IAAUrban/`

Publications

G Google Scholar
† → Equal contribution

Journal Articles

J1. **Yu, Junjie**, Li, W., Li, L., Kumar, D., Sun, H., Topping, D. O., Tang, G. & Zheng, Z. Evaluating the Applicability of ERA5-Land Reanalysis Data for Urban Climate, Health, and Energy Studies in the UK. *Submitted to Urban Climate.*

J2. **Yu, Junjie**, Oleson, K. W., Topping, D. O. & Zheng, Z. Modeling a Rainwater Harvesting and Roof Sprinkling System to Adapt to Urban Extreme Heat. *Submitted to Earth's Future.*

J3. **Yu, Junjie**, Zheng, Z., Ni, J. & Ji, J. Empowerment of accurate modeling of anaerobic membrane bioreactors by automated machine learning. *Journal of Environmental Management* **401,** 128801 (2026).

J4. Xia, J., Ling, F., Li, Z., **Yu, Junjie**, Zhang, H., Topping, D. O., Bai, L. & Zheng, Z. Learning Urban Climate Dynamics via Physics-Guided Urban Surface–Atmosphere Interactions. *Proceedings of the Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025, Main Conference)* (2025).

J5. **Yu, Junjie**, Schreck, J. S., Gagne, D. J., Oleson, K. W., Li, J., Liang, Y., Liao, Q., Sun, M., Topping, D. O. & Zheng, Z. Reinforcement Learning (RL) Meets Urban Climate Modeling: Investigating the Efficacy and Impacts of RL-Based HVAC Control. *ArXiv Preprint* (2025).

J6. **Yu, Junjie**, Sun, Y., Lindley, S., Jay, C., Topping, D. O., Oleson, K. W. & Zheng, Z. Integration and execution of Community Land Model Urban (CLMU) in a containerized environment. *Environmental Modelling & Software* **188,** 106391 (2025).

J7. **Yu, Junjie**, Zheng, Z., Lindley, S., Zhao, L., Wang, C., Wu, Q., Li, L., Topping, D., Schreck, J. S., Gagne, D. J. & Oleson, K. Leveraging Automated Machine Learning (AutoML) for Urban Climate Emulation. *Big Data and Earth System* **1,** 100040 (2025).

J8. Sun, H., Jiao, R., **Yu, Junjie** & Wang, D. Combined effects of particle size and humic acid corona on the aggregation kinetics of nanoplastics in aquatic environments. *Science of the Total Environment* **901** (2023).

J9. Sun, H., Jiao, R., Yang, Q., **Yu, Junjie** & Wang, D. Aggregation and settling characteristics of particulate matter and DOM in a southern China reservoir: Influence of hydraulic conditions and dosing methods. *Process Safety and Environmental Protection* **166,** 500–511 (2022).

J10. Wang, Z., Xu, H., **Yu, Junjie**, Zhao, C. & Wang, D. Effect of particulate matter on coagulation process and ultrafiltration membrane contamination. *Zhongguo Huanjing Kexue/China Environmental Science* **42,** 4621–4630 (2022).

J11. **Yu, Junjie**, Jiao, R., Sun, H., Xu, H., He, Y. & Wang, D. Removal of microorganic pollutants in aquatic environment: The utilization of Fe(VI). *Journal of Environmental Management* **316** (2022).

J12. **Yu, Junjie**, Xu, H., Sun, H., Jin, Z.-Y. & Wang, D.-S. Mechanism on the effects of floc aging and pH adjustment on reflux feed water and coagulation. *Zhongguo Huanjing Kexue/China Environmental Science* **42,** 4612–4620 (2022).

J13. **Yu, Junjie**, Xu, H., Wang, D., Sun, H., Jiao, R., Liu, Y., Jin, Z. & Zhang, S. Variations in NOM during floc aging: Effect of typical Al-based coagulants and different particle sizes. *Water Research* **218** (2022).

J14. **Yu, Junjie**, Xu, H., Yang, X., Sun, H., Jin, Z. & Wang, D. Floc formation and growth during coagulation removing humic acid: Effect of stirring condition. *Separation and Purification Technology* **302** (2022).

J15. Li, M., Xu, H., Wang, D., Wang, X. & **Yu, Junjie**. Comparison of the effect of AlCl3 and Al13 on sludge conditioning in water supply plant. *Chinese Journal of Environmental Engineering* **15,** 1075–1082 (2021).

Presentations

Talks

T1. **Junjie Yu**. *Local Urban Climate-Aware HVAC Control via Reinforcement Learning* Land Model and Biogeochemistry Working Group Meeting 2026. Feb. 2026.

T2. **Junjie Yu**. *Urban Climate Modeling on the AWS.* Urban Weather Intelligence in the Cloud. Jan. 2026.

T3. **Junjie Yu**. *Democratizing Urban Climate Modeling with an Open-Source Framework.* UoM-CUHK joint workshop: Earth system modelling for environmental/climate risk assessments and adaptations. Oct. 2025.

T4. **Junjie Yu**. *Integration and Execution of Community Land Model Urban (CLMU) in a Containerized Environment* 12th International Conference on Urban Climate. July 2025.

T5. **Junjie Yu**. *Integration and Execution of Community Land Model Urban (CLMU) in a Containerized Environment* 30th Annual CESM Workshop. June 2025.

T6. **Junjie Yu**. *Integration and Execution of Community Land Model Urban (CLMU) in a Containerized Environment* Centre for Atmospheric Science seminar. Jan. 2025.

T7. **Junjie Yu**. *Pyclmuapp: A Python Package for Quick Use of Community Land Model Urban.* NERC's Digital Gathering 2024. Aug. 2024.

T8. **Junjie Yu**. *Towards urban climate digital twins: a containerized urban climate model. (Travel funded).* Transforming Urban Living with AI and Digital Twins. Nov. 2024.

T9. **Junjie Yu**. *Reinforcement Learning for Earth and Environmental Sciences.* The Group Meeting of Machine Integration and Learning for Earth Systems of Computational Information Systems Laboratory (CISL) of The National Center for Atmospheric Research (NCAR). Nov. 2023.

Posters

P1. Zhonghua Zheng **Junjie Yu**, K. O. & Zhao, L. *Projections of Global Urban Heat Waves Empowered by Machine Learning.* 4th UK National Climate Impacts Meeting. Aug. 2024.

P2. Kuang Wang **Junjie Yu**, Z. L. & Zheng, Z. *Leverage a Cloud-based Big Data Processing Platform for Climate Extremes Research.* AGU Fall Meeting 2023 Abstract. Dec. 2023.

Teaching

The University of Manchester

2025.08	Instructor, **Urban climate modeling and automated machine learning for weather data modeling** for the exchange program of undergraduate students from the School of Earth Sciences, Zhejiang University, China and the College of Earth Sciences, Jilin University, China
2025.01	Graduate Teaching Assistant, **EART11200 The Natural Scientists Toolkit**
2025.01	Graduate Teaching Assistant, **EART60702 Earth and Environmental Data Science 2023-24 2nd Semester**

2024.08 Instructor, **Observational weather data processing and automated machine learning for weather data modeling** for the exchange program of undergraduate students from the School of Earth Sciences, Zhejiang University, China

Academic Advising

Postgraduates

2025 Mengqi Hu, The University of Manchester (Data Science, School of Social Science, Faculty of Humanities)

Undergraduates

2025 Xinpeng Xu, Zhejiang University (Atmospheric Sciences)
2023 Kuang Wang, Zhejiang University (Biosystems Engineering) -> PhD (Computer Science) at The Chinese University of Hong Kong, Shenzhen

Academic Services

Event Organizations

2026.01 Coordinator of Urban Weather Intelligence in the Cloud, at The University of Manchester, founded by UKRI IAA Starter Fund.
2025.05 Coordinator of Urban Climate Resilience Workshop, at The University of Manchester, founded by UKRI IAA Starter Fund & UoM-CUHK Joint Research Fund.
2025-2026 Coordinator of Seminar of Centre for Atmospheric Science 2025, at The University of Manchester.

Event Supports

2025.03 Assistance of the exchange program of undergraduate students from the School of Environmental Sciences and Engineering, Peking University, China
2024.02 Assistance of the exchange program of undergraduate students from the School of Environmental Sciences and Engineering, Peking University, China

Training

2025 2025 Winter WRF Tutorial (provided by Mesoscale and Microscale Meteorology Laboratory National Center for Atmospheric Research), Online
2025 AWS Workshop on Cloud Research & Teaching Tools, The University of Manchester, Manchester, UK

2023 Training of SQL and cloud computing, Yunqi Academy of Engineering, Hangzhou, China

Skills

1. Programming & AI Frameworks: Python (PyTorch, JAX, PhysicsNeMo), C/C++, Fortran.
2. Machine Learning Expertise: Physics-Informed Neural Networks (PINNs), Generative AI (Diffusion Models), Reinforcement Learning (DQN, PPO), Automated Machine Learning (AutoML).
3. HPC & Data Engineering: Multi-GPU/Distributed Training (CUDA, CuPy), Cloud Computing (AWS ECS), Big Data Storage (Zarr, S3), SQL.
4. Domain Expertise: Earth System Modeling (CESM/CLMU), Regional Weather Modeling (WRF), Spatiotemporal Data Processing.

References

Dr. Zhonghua Zheng

- **Title:** Assistant Professor in Data Science & Environmental Analytics
- **Institution:** Department of Earth and Environmental Sciences, The University of Manchester
- **Email:** zhonghua.zheng@manchester.ac.uk
- **Website:** `https://zhonghuazheng.com/`
- **Relationship:** Supervisor.

Prof. David O. Topping

- **Title:** Professor of The Digital Environment
- **Institution:** Centre for Atmospheric Science, Department of Earth and Environmental Sciences, The University of Manchester
- **Email:** david.topping@manchester.ac.uk
- **Website:** `https://research.manchester.ac.uk/en/persons/david.topping`
- **Relationship:** Supervisor.

Last updated: May 8, 2026